

Spriha Mandal

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Education

Masters in Computer Science (Data Science and Engineering track)

Aug 2023 – May 2025

University of Colorado Boulder (GPA: 3.94/4.0)

Courses: OOP, Data Mining and ML, CV, Data Center Scale Computing, Probability, User-Centered Design and Development.

BTech in Computer Science and Engineering

2016 – 2020

Amrita University (GPA: 9.24/10.0)

Courses: DSA, Networks, OS, DBMS, NLP, Parallel & Distributed Systems, Pattern Recognition in ML, Software Project Mgmt

Experience

AI Engineer, Onki.ai

May 2025 – Present

Technologies: FastAPI, FastHTML, WebSockets, GitHub Actions, Azure, LLM APIs, TTS, Python, JS.

- **CI/CD Pipeline Integration:** Designed and shipped a **GitHub Actions** pipeline with parallel **testing**, linting, and **containerized deploys**, cutting release time and removing manual steps.
- **Automated Documentation Pipeline:** Implemented release-triggered docs and architecture diagram refresh using **LLM**-generated summaries and diagram updates, keeping runbooks and designs current.
- **Real-time LLM Session Orchestration:** Presence-nudge flow (single LLM call on chat history) via new **API + WebSockets**; client-timed **TTS→speech** gaps and **CV** presence signals to interrupt, or automatically end idle sessions.
- **Text-to-Speech Enhancement:** Improving the application's **text-to-speech** capabilities for more natural interactions.

Software Engineering Intern, Rule4, Inc.

Jan 2025 – Apr 2025

Technologies: Django, Django REST Framework, JavaScript, jQuery, js-grid, AJAX, HTML/CSS, CORS, Git

- **Dashboard & APIs:** Built a **Django/DRF** backend (models + REST endpoints) and a **jQuery/js-grid** UI with server-side filtering/pagination; handled cross-browser/**CORS** issues and added solid error handling.
- **Design & Docs:** Wrote API specs and quick docs; added simple validation/tests.

Software Engineer (Chatbot platform), Tata Consultancy Services

Jun 2020 – Dec 2022

Technologies: Node.js, TypeScript, Express/NestJS, Microsoft Bot Framework, WebSockets, Azure (App Service, Functions, Storage), MongoDB, MySQL, TypeORM, ServiceNow Virtual Agent, REST APIs, Docker, Git

- **Core Backend:** Shipped multi-tenant **Node.js/TypeScript** REST/**WebSocket** services on **Azure**; managed session lifecycles, rate limits, and **Bot Framework** channel adapters used in production.
- **ServiceNow Integration:** Delivered the **ServiceNow Virtual Agent** bridge, including **adapters**, **OAuth** setup, and **routing** logic. This enabled linking incidents and requests to internal APIs, ensuring a smooth end-to-end workflow handoff.
- **Ticket Classifier:** Built a lightweight incident-ticket classifier service (feature extraction + model) with a **REST** interface and audit logs, enabling auto-triage and assignment suggestions.
- **Data Layer:** Modeled schemas/repositories with **TypeORM** on **MongoDB/MySQL**; added migrations, indexes, and pooling.
- **Reliability:** Added retries/backoff, idempotent endpoints, health checks, and graceful shutdowns for APIs/workers.
- **Security:** Helped implement JWT/OAuth, role-based access, input validation, and request limits.
- **Delivery & Observability:** Dockerized services, contributed to **Azure** CI/CD, and added structured logs for debugging.

Projects

VoxFormat (Speech-to-Formatted_text) [🔗](#)

- Built a **C++20** CLI with **multi-threaded** audio capture/processing; **PortAudio** mic input, **libsamplerate** resampling, streamed to **whisper.cpp** for real-time transcription (**CMake** build).
- Added **voice-driven Markdown formatting**, live terminal preview, and silence-based auto-stop for clean session endings.

IIoT Demonstration Town project (Masters Capstone project)

- Implemented a secure **React/Node.js** HMI over Modbus TCP to control **PLC** devices.
- Configured **Ubuntu Server 24.04** networking with **Netplan**, TLS 1.2, and multi-client WPA2 Wi-Fi/Ethernet.

NLP from scratch - Named Entity Recognition

- Developed a full pipeline for scientific entity recognition in research papers, covering data processing, annotation, model training, and deployment using **Python**, **PyTorch**, **HuggingFace**, **PySpark**, and **AWS**.
- Fine-tuned **RoBERTa** and **DeSciBERTa** models for sequence tagging with domain-specific pretraining, achieving an **F1 score of 0.82** on a private scientific NER dataset.

Achievements

- Among top **100 out of 7276 applicants** from India for **Google Women Techmakers Engineering Fellowship 2019**.
- **Semi-finalist** in university-sponsored hackathon **CU Hacks 2024**.
- The only girl to get selected from my entire campus for **Amazon Campus Mentorship Series 2019**.
- **Semi-finalist out of 267,000+** participants in **TechGig Code Gladiators 2019 coding competition**.